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Stochastic Modelling of Periodicity of Infectious Disease

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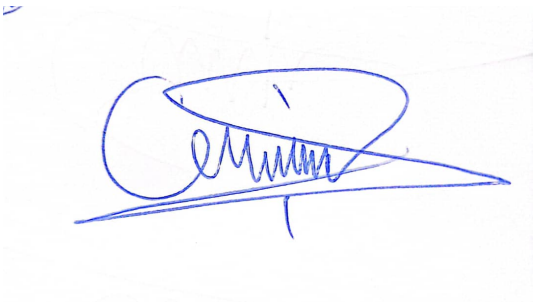
Abstract

This thesis investigates the stochastic modeling of the periodicity of infectious diseases through a compartmental model named as Susceptible-Infectious-Immunized-Declining Immunity (SIRWS), which integrates three fundamental biological mechanisms. These biological mechanisms are seasonal forcing of the transmission rate, progressive loss of immunity in two stages (waning), and immune restoration through re-exposure (boosting). Building upon the deterministic framework and the seasonal forcing approach of previous research, this work extends the analysis to the stochastic domain by formulating the SIRWS model as a non-homogeneous continuous-time Markov chain (CTMC). In the deterministic framework, the basic reproduction number R_0 is calculated using the next-generation matrix method, and the instantaneous reproduction number $R(t_0)$ is defined to capture the intra-annual variation in transmission capacity. Numerical simulations show that seasonal forcing does not generate epidemic cycles but amplifies, regularizes, and synchronizes the endogenous oscillations produced by the waning-boosting mechanism. In the stochastic framework, the model is formalized using infinitesimal transition probabilities, and the forward and backward Kolmogorov equations are rigorously derived. Using the probability generating function, we analytically establish the expected number and variance of infected individuals. The rapid growth of the variance reveals that epidemic uncertainty is an intrinsic property of the stochastic process. The extinction probability $P_0(t)$ oscillates seasonally, identifying winter as the most dangerous period for the introduction of a pathogen. Exact stochastic trajectories are simulated using Gillespie's algorithm and compared to the deterministic ODE solution. Three phenomena absent from the deterministic framework are demonstrated: stochastic extinction, stochastic phase shift, and inter-trajectory variability. The central original contribution of this thesis is the first stochastic comparison of the SIRWS and SIRS models, revealing that immune boosting acts as a stochastic stabilizer by lengthening inter-epidemic cycles and reducing trajectory variability. This result extends to the stochastic framework the work based on deterministic models by researchers.

Keywords: SIRWS model, CTMC, Gillespie algorithm, seasonal forcing, immune boosting, extinction probability, epidemic periodicity, pertussis, Kolmogorov equations.

Declaration

I, the undersigned, hereby declare that the work contained in this essay is my original work, and that any work done by others or by myself previously has been acknowledged and referenced accordingly.



Ulrich Valdes TCHOMBA NGOUNOU, 23 June 2026.

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1. Introduction

1.1 Background of the study

Infectious diseases have long been one of the most significant threats to global public health, causing millions of deaths each year and placing a considerable burden on health systems, particularly in low- and middle-income countries. According to the World Health Organization's 2004 Global Burden of Disease report, infectious and parasitic diseases caused approximately 9.5 million deaths in 2004, representing 16.2% of global mortality [17]. Despite remarkable advances in medicine, vaccination, and public health infrastructure over the past century, many infectious diseases continue to circulate endemically in human populations and periodically re-emerge in epidemic waves.

A particularly striking and well-documented characteristic of many infectious diseases is their tendency to exhibit periodic, recurring epidemics. Before the introduction of widespread vaccination, measles exhibited remarkably regular biennial epidemic cycles in large urban populations such as London and New York [3, 8]. Influenza returns annually during the winter season in temperate climates, influenced by seasonal fluctuations in temperature, humidity, and human contact patterns [14]. Pertussis has historically exhibited cycles of 3 to 4 years, and its recent resurgence in highly vaccinated populations has renewed scientific interest in the mechanisms underlying its periodicity [12]. Cholera, caused by the bacterium Vibrio cholerae, has caused seven pandemics since 1817, and continues to affect between 1.3 and 4 million people each year, with an estimated 21,000 to 143,000 annual deaths [WHO, 2017, *https://punchng.com/worlds-worst-cholera-outbreak-in-yemen/*].

The periodic nature of these epidemics is not simply due to chance. It reflects a complex interaction between several biological and environmental mechanisms. First, the depletion and renewal of the susceptible population through births, deaths, and loss of immunity play a fundamental role in determining the inter-epidemic interval [3]. Second, seasonal forcing of the transmission rate linked to climatic variables such as temperature and humidity, as well as social factors like the start of the school year and contact behaviors, introduces a periodic external perturbation into the system that can drive, amplify, or complicate endogenous oscillatory dynamics [8]. Third, the duration and maintenance of immunity after infection or vaccination are critical determinants of disease dynamics. For diseases such as pertussis, immunity is not permanent but declines gradually over time and can be partially restored by re-exposure to the circulating pathogen. This is a phenomenon known as immune boosting [6, 12].

Mathematical modeling has become an indispensable tool for understanding, analyzing, and predicting these complex epidemic dynamics. Since the seminal work of Kermack and McKendrick [11], who introduced the SIR compartmentalized framework and established the threshold theorem for epidemic invasion, the field of mathematical epidemiology has developed considerably. Deterministic ordinary differential equation (ODE) models have been widely used to study the qualitative behavior of epidemic systems, including the existence and stability of equilibria, the calculation of the basic reproduction number R_0, and the conditions under which periodic solutions can emerge via Hopf bifurcations [5, 6]. However, deterministic models inherently ignore the randomness of disease transmission. In reality, each infection event results from random contact between an infectious individual and a susceptible individual, and this stochasticity can have significant consequences for the dynamics of the system, particularly in small or medium-sized populations. Stochastic models, based on Continuous-Time Markov Chains (CTMCs) and stochastic differential equations (SDEs), provide a more realistic framework by explicitly incorporating this element of randomness [2, 15]. They allow for the calculation of extinction probabilities, the analysis of noise-induced oscillations, and the study of how demographic stochasticity

modifies the periodic patterns predicted by deterministic models. Essentially, stochastics explains that a disease can disappear simply by chance.

1.2 Statement of the problem

The vast majority of mathematical studies on the periodicity of infectious diseases rely on deterministic models [4, 16], in which the evolution of the system is entirely predictable for given initial conditions. However, the transmission of infectious diseases is inherently stochastic: contact between an infectious individual and a susceptible individual is a random event, transmission rates vary unpredictably, and small populations can experience epidemics that die out purely by chance, even when the basic reproduction number R_0 is greater than 1. Moreover, the immunity conferred by infection is not permanent: it declines over time (waning immunity) and can be restored by re-exposure to the pathogen (immune boosting), two mechanisms that further complicate the dynamics. In this context, it is crucial to understand how the seasonal forcing of the transmission rate interacts with demographic stochasticity and the mechanisms of waning and boosting immunity to generate the periodicity patterns observed in real-world epidemiological data.

1.3 Objectives of the study

This thesis pursues the following general and specific objectives.

1.3.1 General objective. The general objective of this study is to analyze how seasonal forcing affects the periodicity of epidemics in a SIRWS model with decaying immunity and immune boosting, extending the analysis from the deterministic to the stochastic framework, more precisely the continuous time markov chain (CTMC).

1.3.2 Specific objectives:. The specific objectives of this study are

- To present and analyze the deterministic SIRWS model with a constant transmission rate, calculate the basic reproduction number R_0, and study the stability of the equilibria,
- To introduce a periodic transmission rate $\beta(t)$ and analyze its impact on the model dynamics: forced oscillations, bifurcations, and the possible emergence of chaos,
- To construct the stochastic extension of the SIRWS and SIRS models as a continuous time markov chain (CTMC) and to compare them, and
- To compare the deterministic and stochastic dynamics, in particular the impact of stochasticity on the periodicity of epidemics.

1.4 Significance of the study

This work has dual relevance, both scientific and practical. From a scientific perspective, it contributes to the understanding of the endogenous and exogenous mechanisms that generate epidemic periodicity, by combining three elements rarely addressed together in the literature: progressive loss of immunity,

immune boosting, and seasonal forcing, within a rigorous stochastic framework, more precisely the ontinuous time tarkov chain (CTMC). From a practical perspective, the results of this work can contribute to improving epidemic forecasting and better assessing the impact of public health interventions, such as vaccination campaigns, on the dynamics of diseases with non-permanent immunity. This work is particularly relevant for diseases like pertussis (a strong periodicity of 3 to 4 years, one of the diseases with the highest basic reproduction number $R_0 \in [12, 17]$, combining all three: periodicity, loss of immunity, and natural boosting), whose recent resurgence in several countries, despite high vaccination coverage, constitutes a major public health problem.

1.5 Structure of the thesis

This thesis is organized into four chapters. Chapter 1 constitutes the general introduction and presents the context, the problem statement, the objectives, and the relevance of the study. Chapter 2 is devoted to the literature review, covering the historical evolution of infectious disease modeling, compartmental models, epidemic periodicity, and stochastic approaches. Chapter 3 presents the analysis of the deterministic SIRWS model, first with a constant transmission rate and then with a seasonal rate, including the calculation of R_0, stability analysis, and numerical simulations. Chapter 4 develops the stochastic extension of the model in the form of a continuous time markov chain (CTMC), presents the Gillespie algorithm, and compares the deterministic and stochastic dynamics.

2. Literature review

During the first half of the twentieth century, humanity remained vulnerable to epidemics: the 1918 flu claimed more than 50 million lives, disrupting demographics, paralyzing trade, and drastically restricting freedom of movement. These health crises demonstrated that a disease is not merely a medical problem: it alters birth rates, migration patterns, social behaviors, and the economy. Understanding these dynamics to better control them thus became a major scientific challenge. It was to meet this challenge that researchers from mathematics and medicine joined forces. While mathematical modeling of epidemics began as early as 1760 with the mathematician Daniel Bernoulli's work on smallpox, it was in 1927 with the biochemist W. O. Kermack and the military physician A. G. McKendrick, both British researchers, that modern epidemiology truly emerged [11]. In their article "A Contribution to the Mathematical Theory of Epidemics," they synthesized: the deterministic compartmental SIR structure, the epidemic threshold, simple equations, and a proper mathematical analysis.

Since then, several epidemic models have been analyzed. In order to bring the study of predictions closer to reality, researchers have taken into account the periodicity of the seasons. Thus, in the year 2000, researchers D.Earn and B. Bolker [8] introduced seasonal variations into a deterministic SIR model for the first time. In their research, they show that the dynamic transitions observed for measles in the 20th century can be explained by the combination of a declining birth rate and seasonal forcing.

With the aim of understanding the true workings of infectious diseases and improving their prevention and control, researchers are considering the permanent or temporary immunity conferred by certain diseases, and the possibility of boosting it in the event of reinfection. In 2011, Lavine [12] introduced the SIRWS model with the immune boosting mechanism to explain the resurgence of pertussis despite high vaccination coverage. Dafilis et.al [5] went further in 2014 by making a direct extension. They showed that the combination of seasonality, boosting, and birth rate generates extremely rich dynamics: whole-period cycles, Arnold's tongues, period doublings, and chaos.

Deterministic models have been the primary tool for understanding, anticipating, and combating infectious diseases. However, these models essentially ignore the random nature of disease transmission and sometimes predict outcomes in advance. In reality, every infection results from random contact between an infected individual and a susceptible individual, and this stochasticity can have significant consequences for the dynamics of the system, particularly in small or medium-sized populations.

The earliest, or perhaps the very first, article that explores the interaction between determinism and stochasticity in childhood diseases was published in 2002 by Rohani and Grenfell [15]. In their article, they show that stochastic noise can maintain oscillations in systems that would deterministically converge to equilibrium and can induce local extinctions even when $R_0 > 1$.

Several other works on stochastic models in epidemiology have been published, notably Allen's books from 2008 and 2017 [1, 2]. These books present birth-death processes, continuous-time Markov chains, and the Gillespie algorithm in a clear and rigorous manner.

One of the most recent articles in stochastic epidemiology relevant to this work is that of Hridoy [10], published in 2024. In this article, the researcher provides a comprehensive review of modeling approaches for capturing seasonal transmission in stochastic epidemic models, with a particular focus on continuous-time Markov chains (CTMCs) and branching processes. Using an SIR model, he performs a systematic comparison of five seasonal forcing functions, and the analysis highlights the "winter is coming" effect, where the probability of a major outbreak peaks just before the seasonal transmission maximum. This work is methodologically closest to ours, as it employs the same stochastic tools as Gillespie simulations

and extinction probability analysis, though without waning immunity or boosting.

Despite significant advances in understanding epidemic periodicity, few studies have jointly examined the effect of seasonal forcing and demographic stochasticity in a model integrating both progressive loss of immunity and immune boosting. Dafilis et al. [5] studied the SIRWS model with seasonality but within a purely deterministic framework. Rohani et al. [15] studied stochasticity but without waning immunity or boosting. This thesis aims to fill this gap by extending the seasonal SIRWS model of Dafilis et al. [5] to a stochastic framework based on continuous-time counting processes.

3. Deterministic model formulation and analysis

In this Chapter, we formulated a deterministic model of infectious disease and studied the basic mathematical properties of the model.

3.1 Model assumptions

The primary assumption of our deterministic model development is, we divide the whole population into four compartments. These are susceptible (S) who are healthy individuals at risk of infection, wnfectious (I)who are individuals capable of transmitting the disease, recovered (R) who are individuals who have cleared the infection and acquired full immunity, and waning immunity (W) who are individuals whose immunity is partially developed but not completely susceptible like those in S. Assuming N denote the total population size over time, we have

$$N = S(t) + I(t) + R(t) + W(t) \quad \forall t \geq 0. \tag{3.1.1}$$

The other assumptions of model development are given as follows.

A1: Homogeneous population: There is no age structure, no spatial structure, and no individual heterogeneity. All individuals in the same compartment are identical and interchangeable.

A2: Closed population: there is no migration. Births are directly offset by the death rate.

A3: Mortality independent of the disease: The disease does not cause death, but there is natural death in each compartment with rate μ.

A4: The transmission rate is first considered constant equal β, then later time-dependent $\beta(t)$ to account for seasonality.

A5: No latency: when susceptible people meet an infected individual, they become infectious immediately.

A6: Only infected (I) transmit the disease: there is no disease transmission from any compartment other than (I).

A7: No permanent immunity: immunity wanes after an average time $T_w = \frac{1}{\kappa_1} + \frac{1}{\kappa_2}$ after infection. Indeed, recovered individuals (R) gradually lose immunity, and transition first from compartment (R) to compartment (W) with a rate κ_1,representing individuals with waning or partial immunity. Immunity is then lost completely, and individuals return to the susceptible compartment (S) with the rate κ_2.

A8: Possible boosting: a person in (W) can be boosted by even minimal re-exposure and return to R, with rate of boosting equal η.

Based on the assumptions, we obtain the flow SIRWS diagram in Figure 3.1.

For better organization and clarity of the model development, description of model state variables and model parameters are presented in Table 3.1 and 3.2, respectively.

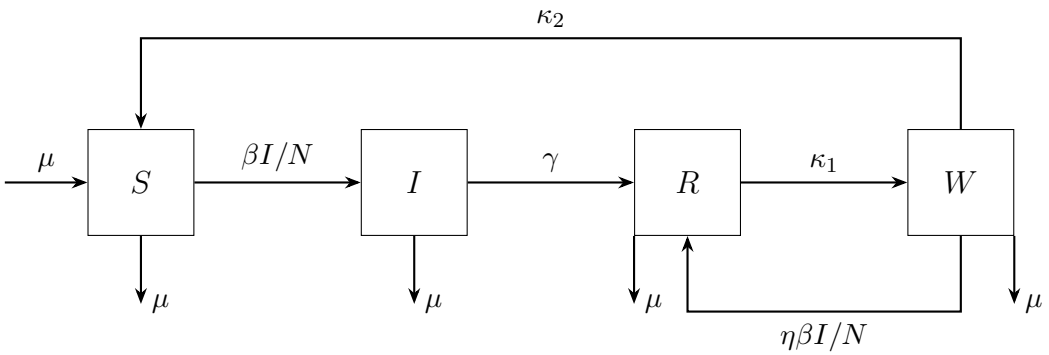


Figure 3.1: Flow diagram of the model.

Table 3.1: Description of variables

Variables	Names	Description
S	Susceptibles	Individuals at risk of infection
I	Infectious	Contagious Individuals
R	Recovered	individuals who have cleared the infection and acquired full
W	Waning immunity	individuals whose immunity wanes but not completely susceptible like in S.

Table 3.2: Description of parameters

Parameters	Names	Unit	Interpretation
β	Transmission rate	1/(person $\times$day)	$\beta = c \times p$; $c =$ contacts/day,$p =$transmission proba
γ	Recovered rate	1/day	$1/\gamma =$ average duration of infection
κ_1	partial waning rate	1/day	$1/\kappa_1 =$ average duration of immunity
κ_2	total lost immunity rate	1/day	$1/\kappa_2 =$ average duration of immunity
η	boosting rate	1/day	rate at which a W becomes fully immune
μ	birth/death rate	1/day	$1/\mu =$ life expectancy

The model is defined by a non-autonomous system of ordinary differential equations (ODEs). Using the flow diagram and assumptions, the desired mathematical model is given as

$$\begin{aligned}
\frac{dS}{dt} &= \mu N - \frac{\beta(t)I}{N}S + \kappa_2 W - \mu S,\\
\frac{dI}{dt} &= \frac{\beta(t)I}{N}S - \gamma I - \mu I,\\
\frac{dR}{dt} &= \gamma I - \kappa_1 R - \mu R + \eta\frac{\beta(t)I}{N}W,\\
\frac{dW}{dt} &= \kappa_1 R - \mu W - \kappa_2 W - \eta\frac{\beta(t)I}{N}W.
\end{aligned} \tag{3.1.2}$$

In Equation 3.1.2, the transmission rate $\beta := \beta(t)$ is taken to be

$$\beta(t) = \beta_0\left(1 + \varepsilon\cos\left(\frac{2\pi t}{\omega} + \phi\right)\right), \tag{3.1.3}$$

where t is measured in days is the length of the cycle, β_0 is the baseline transmission rate around which seasonal fluctuations occur, $\varepsilon \in [0, 1]$ is the amplitude of the seasonal forcing, ω represents the period of

the seasonal cycle (for instance, 12 months for an annual cycle) and ϕ is the phase shift that determines the month of peak transmission. This form is justified by Hridoy [10] as the most appropriate for diseases with a regular annual cycle such as pertussis or influenza.

3.2 Basic properties of a time-independent model

To provide a more detailed analysis, we first consider the model with a constant transmission rate ($\varepsilon = 0$). Subsequently, we incorporate seasonal transmission ($\varepsilon \neq 0$) to better assess its impact on the system dynamics. By considering $\varepsilon = 0$, the model with constant transmission rate is given by the following autonomous system:

$$\begin{aligned}
\frac{dS}{dt} &= \mu N - \frac{\beta_0 I}{N} S + \kappa_2 W - \mu S, \\
\frac{dI}{dt} &= \frac{\beta_0 I}{N} S - \gamma I - \mu I, \\
\frac{dR}{dt} &= \gamma I - \kappa_1 R - \mu R + \eta \frac{\beta_0 I}{N} W, \\
\frac{dW}{dt} &= \kappa_1 R - \mu W - \kappa_2 W - \eta \frac{\beta_0 I}{N} W.
\end{aligned} \tag{3.2.1}$$

3.2.1 Proposition. (Existence and uniqueness of solution)

Given non-negative initial conditions $S(0), I(0), R(0), W(0)$, the system in Equation (3.2.1) has a unique solution.

Proof. Let $X = (S, I, R, W)$ and define $F(X) = (F_S(X), F_I(X), F_R(X), F_W(X))$ where:

$$\begin{aligned}
F_S(X) &= \mu N - \frac{\beta_0 I}{N} S + \kappa_2 W - \mu S, \\
F_I(X) &= \frac{\beta_0 I}{N} S - \gamma I - \mu I, \\
F_R(X) &= \gamma I - \kappa_1 R - \mu R + \eta \frac{\beta_0 I}{N} W, \\
F_W(X) &= \kappa_1 R - \mu W - \kappa_2 W - \eta \frac{\beta_0 I}{N} W.
\end{aligned}$$

The system in Equation (3.2.1) can be written as $\frac{dX}{dt} = F(X)$ with $X(0) = (S(0), I(0), R(0), W(0))$.

F is of class $C^1(\mathbb{R}^4)$ (indeed, all its component are). Hence, by the Cauchy-Lipschitz theorem, there exists a unique solution for given initial conditions. □

3.2.2 Proposition. (Feasibility region of the model)

The subset of $\mathbb{R}^4_+$ given by

$$\mathcal{T} = \left\{ (S, I, R, W) \in \mathbb{R}^4 : 0 \leq S + I + R + W \leq N \right\},$$

is positively invariant for the system. That is, any trajectory starting in $\mathcal{T}$ remains in $\mathcal{T}$ for all $t \geq 0$.

Proof. The proof proceeds in two steps: we first show that all compartments remain non-negative, and then we show that the total population never exceeds N.

Step 1: Non-negativity of solutions. We apply the barrier theorem: if a variable reaches zero and its derivative at that point is non-negative, it cannot cross below zero.

For S: if $S(t_1) = 0$, then

$$\frac{dS}{dt}(t_1) = \mu N + \kappa_2 W(t_1) \geq 0,$$

since $\mu N > 0$ and $W(t_1) \geq 0$. Thus S cannot become negative.

For I: if $I(t_1) = 0$, then every term in $\frac{dI}{dt}$ contains a factor of I, so

$$\frac{dI}{dt}(t_1) = 0.$$

Hence I remains non-negative.

For R: if $R(t_1) = 0$, then

$$\frac{dR}{dt}(t_1) = \gamma I(t_1) + \eta \frac{\beta_0 I(t_1)}{N} W(t_1) \geq 0,$$

because $I(t_1), W(t_1) \geq 0$. So R cannot become negative.

For W: if $W(t_1) = 0$, then

$$\frac{dW}{dt}(t_1) = \kappa_1 R(t_1) \geq 0,$$

because $\kappa_1 > 0$ and $R(t_1) \geq 0$. Thus W remains non-negative.

Therefore, all four compartments stay non-negative for all $t \geq 0$.

Step 2: Upper bound. Let $T(t) = S(t) + I(t) + R(t) + W(t)$. Summing the four equations, all interaction terms cancel, leaving

$$\frac{dT}{dt} = \mu N - \mu T = \mu(N - T).$$

Solving this linear ODE gives

$$T(t) = N - (N - T(0))e^{-\mu t}.$$

Since $T(0) \leq N$ by hypothesis, we have $T(t) \leq N$ for all $t \geq 0$.

Consequently, $\mathcal{T}$ is positively invariant. □

3.3 Disease free equilibrium and basic reproduction number

The basic reproduction number R_0 is the average number of new infections generated by an infected individual in a completely susceptible population during his/her entire infectious period.

3.3.1 Proposition. (Disease-free equilibrium (DFE))

The point $E_0 = (N, 0, 0, 0)$ is the disease free equilibrium for the system in Equation (3.2.1).

Proof. At the disease-free equilibrium, we consider the absence of disease:

$$I = R = W = 0.$$

From the system in Equation (3.2.1), the change in the susceptible population becomes

$$\frac{dS}{dt} = 0 \implies \mu(N - S) = 0 \implies S = N.$$

Thus, the disease-free equilibrium is given as

$$E_0 = (S^0, I^0, R^0, W^0) = (N, 0, 0, 0).$$

□

3.3.2 Proposition. (Basic reproduction number)

The basic reproduction number of model system in Equation (3.2.1) is given by

$$R_0 = \frac{\beta_0}{\gamma + \mu}. \tag{3.3.1}$$

Proof. The only infected compartment is I. New infections arise from the mass action term $\beta_0 IS/N$. Using the next-generation matrix algorithm, the vector of new infections is $\mathcal{F} = \beta IS/N$ and the vector of transitions is $\mathcal{V} = (\gamma + \mu)I$. Evaluating these two matrices at the DFE, we found that

$$F = \left.\frac{\partial \mathcal{F}}{\partial I}\right|_{\text{DFE}} = \frac{\beta_0 \cdot S^0}{N} = \frac{\beta_0 \cdot N}{N} = \beta_0, \qquad V = \left.\frac{\partial \mathcal{V}}{\partial I}\right|_{\text{DFE}} = \gamma + \mu.$$

The basic reproduction number is the spectral radius of the next-generation matrix FV^{-1} and is derived to be

$$R_0 = \rho(FV^{-1}) = \frac{\beta_0}{\gamma + \mu}.$$

□

This basic reproduction number tells us that

- If $R_0 < 1$, disease decays (extinction),
- If $R_0 > 1$, disease grows (epidemic), and
- If $R_0 = 1$ is the epidemic threshold.

3.3.3 Analysis of seasonnality: SIRWS-Model with periodic transmission rate ($\varepsilon \neq 0$). The objective of this subsection is to analyze the effect of seasonality on the dynamics of the SIRWS system, in order to understand how periodic variations in the transmission rate (seasonal forcing) enrich the observed behaviors. Then we reconsider the non-autonomous system in Equation (3.1.2) and we aim to examine how the seasonal amplitude ε and the shift ϕ modifies the behavior of the endemic equilibrium, the nature and amplitude of oscillations, and how it can generate more complex phenomena such as period-doubling or chaos.

In this work, we introduce seasonal forcing into the transmission rate $\beta(t) = \beta_0 \left(1 + \varepsilon \cos\left(\frac{2\pi t}{\omega} + \phi\right)\right)$, with $0 \leq \epsilon \leq 1$ representing the seasonality intensity.

3.3.4 Definition (Instant reproduction number)**.** The instant reproduction number of system in Equation (3.1.2) at time t_0 is defined by

$$R(t_0) = \frac{\beta(t_0)}{\gamma + \mu} = \frac{\beta_0 \left(1 + \varepsilon \cos\left(\frac{2\pi t_0}{\omega} + \phi\right)\right)}{\gamma + \mu}.$$

However, this $R(t_0)$ does not have the same interpretation than the R_0 define in Equation (3.3.1).

3.4 Reproduction number with time-dependent transmission

In the previous section, the basic reproduction number $R_0 = \beta_0/(\gamma + \mu)$ was computed using the next-generation matrix method. This number measures the average spreading capacity of the disease over a full year. However, in a model with seasonal transmission rate $\beta(t)$, this capacity varies throughout the year. It is therefore natural to define a time-dependent reproduction number $R_0(t)$ that measures this capacity at a any specific time t.

3.4.1 General formula of basic reproduction number. In general, the basic reproduction number is define as [7]

$$R_0 = \int_0^{+\infty} \beta(t)\theta(t)dt, \tag{3.4.1}$$

where $\theta(t)$ is the survival function of an individual in the infectious state, that is, the probability that an individual is still infectious at time t after having been infected.

Under the standard hypothesis of compartmental models, the time spent under infection follows an exponential distribution. Given that an individual leaves the compartment I at the rate $\gamma + \mu$ (either through recovery or natural death), then the probability of still being infectious at time t is given by:

$$\theta(t) = e^{-(\gamma+\mu)t}, \quad t > 0.$$

Substituting into Equation (3.4.1), we obtain:

$$R_0 = \int_0^{+\infty} \beta_0 \left(1 + \varepsilon \cos\left(\frac{2\pi t}{\omega} + \phi\right)\right) e^{-(\gamma+\mu)t} dt. \tag{3.4.2}$$

Separating this integral, we obtain

$$R_0 = \beta_0 \int_0^{+\infty} e^{-(\gamma+\mu)t} + \beta_0 \varepsilon \int_0^{+\infty} \cos(\omega_0 t + \phi) e^{-at} dt, \tag{3.4.3}$$

with $a = \gamma + \mu \quad and \quad \omega_0 = \frac{2\pi}{\omega}$. For the first part, we have

$$\beta_0 \int_0^{+\infty} e^{-(\gamma+\mu)t} dt = \frac{\beta_0}{\gamma + \mu}.$$

For the second hand, using double integration by parts, we obtain

$$\begin{aligned}\int_0^{+\infty} \cos(\omega_0 t+\phi)e^{-at}dt &= \frac{a\cos(\phi)-\omega_0\sin(\phi)}{a^2+\omega_0^2},\\ &= \frac{(\gamma+\mu)\cos(\phi)-\frac{2\pi}{\omega}\sin(\phi)}{a^2+(\frac{2\pi}{\omega})^2},\\ &= \frac{\omega^2(\gamma+\mu)\cos(\phi)-2\pi\omega\sin(\phi)}{\omega^2(\gamma+\mu)^2+4\pi^2}.\end{aligned}$$

Substituting these two expressions into Equation (3.4.3), we obtain:

$$R_0 = \beta_0\left[\frac{1}{\gamma+\mu}+\varepsilon\frac{\omega^2(\gamma+\mu)\cos(\phi)-2\pi\omega\sin(\phi)}{\omega^2(\gamma+\mu)^2+4\pi^2}\right]. \tag{3.4.4}$$

This constant can be used in the same way as the previous R_0 to identify supercritical $(R_0>1)$ and subcritical $(R_0<1)$ processes.

3.4.2 Dynamics of $I(t)$ near the DFE with constant β. Near the disease-free equilibrium (DFE), $S\approx N$. With constant transmission rate β_0 $(\varepsilon=0)$, the equation for I in the system (3.2.1) reduces to

$$\frac{dI}{dt}\approx\big[\beta_0-(\gamma+\mu)\big]I.$$

This is a separable linear ODE. Using separation of variables and integrating from 0 to t, we get that

$$\ln\left(\frac{I(t)}{I(0)}\right)=(\beta_0-(\gamma+\mu))t.$$

The above equation can be written as

$$I(t)=I(0)\,\exp\Big((\gamma+\mu)(R_0-1)t\Big), \tag{3.4.5}$$

where $R_0=\beta_0/(\gamma+\mu)$.

3.4.3 Dynamics of $I(t)$ with seasonal $\beta(t)$. Near the disease-free equilibrium (DFE), $S\approx N$. With time dependent transmission rate $\beta(t)$ $(\varepsilon\neq 0)$ given in Equation (3.1.3), the equation for I in the system (3.2.1) reduces to

$$\frac{dI}{dt}\approx\big[\beta(t)-(\gamma+\mu)\big]I.$$

The coefficient is no longer constant, it depends on time. We separate the variables again and we integrate from 0 to t so that

$$\ln\left(\frac{I(t)}{I(0)}\right)=\int_0^t\beta(s)ds-\big(\gamma+\mu\big)t. \tag{3.4.6}$$

From the above equation, we get that

$$I(t) = I(0)\,\exp\left[(\gamma+\mu)\left(\frac{\int_0^t \beta(s)ds}{(\gamma+\mu)t} - 1\right)t\right].$$

Let us define a time-dependent basic reproduction number at $t \geq 0$ as

$$R_0(t) = \frac{\int_0^t \beta(s)ds}{(\gamma+\mu)t}. \tag{3.4.7}$$

Thus, the number of infected is given as

$$I(t) = I(0)\,\exp\Big[(\gamma+\mu)\,(R_0(t)-1)\,t\Big]. \tag{3.4.8}$$

It is clear, from Equation (3.4.6) that the epidemic goes to extinction if

$$\lim_{t\to+\infty} \int_0^t \beta(s)ds - (\gamma+\mu)t = -\infty, \tag{3.4.9}$$

or equivalently $\lim_{t\to+\infty} R_0(t) < 1$. Similar argument indicates that $\lim_{t\to+\infty} R_0(t) > 1$ leads to a supercritical process.

3.5 Numerical simulations

3.5.1 Simulation setup. Numerical simulations are performed in Python, using the built-in odeint function from the SciPy library, which implements the LSODA solver with automatic time step control. Table (3.3) lists the parameter values used, unless otherwise stated.

Table 3.3: Summary of model parameters. Time unit is one day [6, 10, 13].

Parameter	Numerical value	Biological meaning
β_0	$30/365$	An infectious infects 30 people per year in a susceptible population
γ	$17/365$	Recovery occurs after approximately 21 days on average
μ	$1/(80\times 365)$	Natural death occurs after approximately 80 years on average
ε	0.3	$\beta(t)$ varies between 70% and 130% of β_0 depending on season
κ_1	$1/365$	Partial loss of immunity: $R \to W$ after approximately 1 year
κ_2	$0.5/365$	Total loss of immunity: $W \to S$ after approximately 2 years
η	4.0	rate at which a waning individual recovers full immunity
ϕ	0.0 rad	Transmission peak on January 1st
ω	365 days	Complete annual cycle
N	1000	Total population

3.5.2 Variation of $\beta(t)$ and $R(t)$ over one year. This shows the seasonality of the transmission rate and the instantaneous reproduction number. It explains why epidemics have peaks in winter.

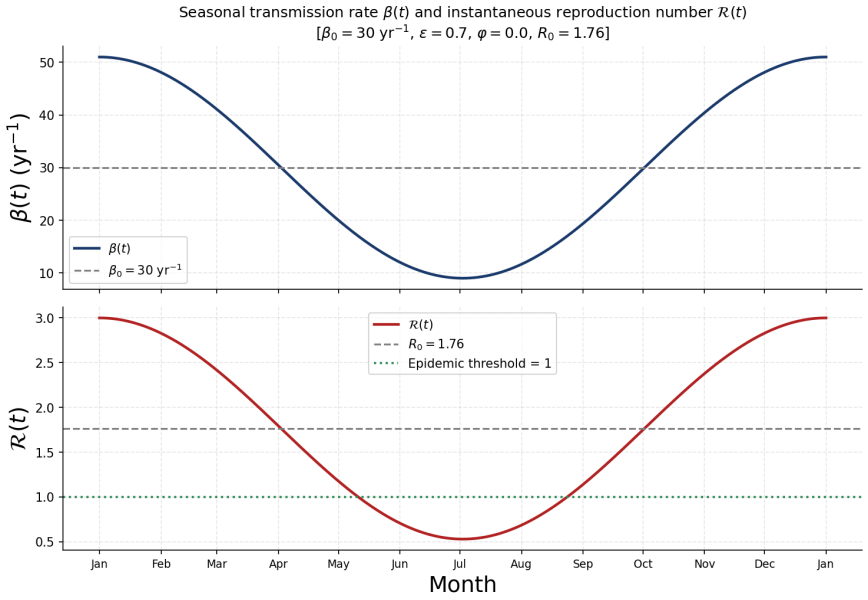


Figure 3.2: Seasonal transmission rate $\beta(t)$ and instantaneous reproduction number $\mathcal{R}(t)$ over one year. $\beta(t)$ oscillates and $R(t)$ with period $\omega = 365$ days and the phase shift $\phi = 0$ ensures the peak occurs at the beginning of the year.

Figure 3.2 shows how the seasonality transmission rate and the time dependent basic reproduction number behave over time. Seasonality induces a significant variation in the transmission rate. The value $\varepsilon = 0.3$ means that $\beta(t)$ varies by $\pm 30\%$ around β_0. In winter (January), transmission is at its maximum because contacts are more frequent. In summer (July), it drops to its minimum. With this variation of $\beta(t)$, $R(t)$ falls below 1 in summer (between May and September). This means that transmission is insufficient to sustain the epidemic during this period. Consequently, it dies out or decreases sharply. Each winter (between January and April, then October and December), $R(t)$ rises above 1 again, which triggers a new epidemic wave if susceptible individuals are available.

3.5.3 Deterministic SIRWS simulation: $\varepsilon = 0$ vs $\varepsilon \neq 0$ over Ten years. The plots in Figure 3.3 show the difference between the model without seasonality (constant β) and with seasonality ($\beta(t)$).

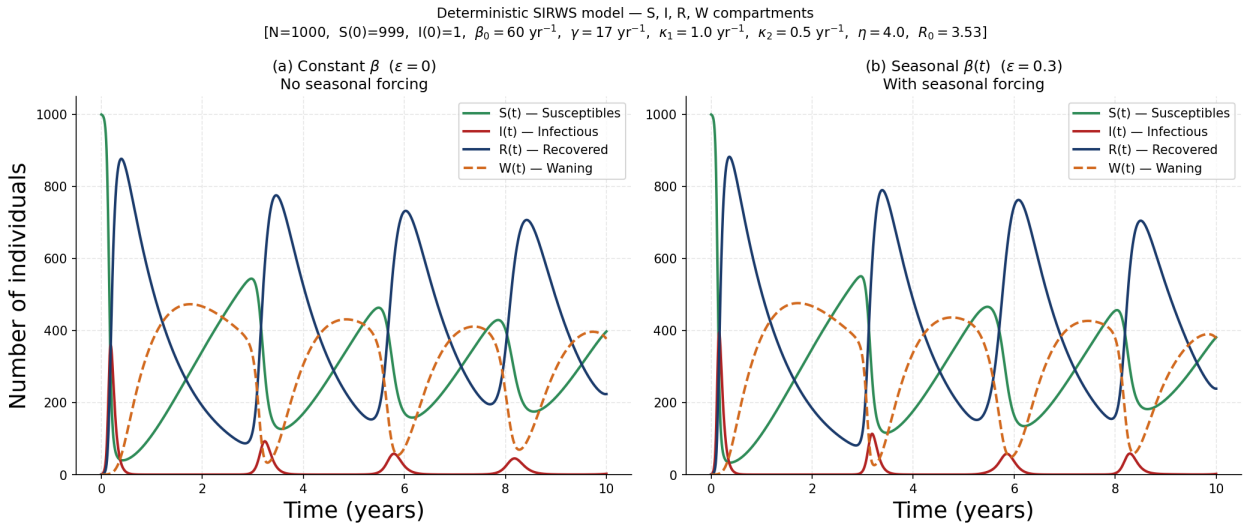


Figure 3.3: Trajectories of the different populations in the SIRWS model with and unless seasonality effect over a 10-year period. Baseline transmission rate $\beta_0 = 60/365$, $S_0 = 999$, I_0, duration of immunity of 3 years with $\kappa_1 = 1/365$, $\kappa_2 = 0.5/365$.

The SIRWS model generates epidemic oscillations even in the absence of seasonal forcing ($\varepsilon = 0$), as shown in Figure 3.3(a). These oscillations called endogenous oscillations, are a direct consequence of the mechanism of progressive loss of immunity (3-year immunity that gradually declines :$1/\kappa_1 = 1$ year for the partial decline from R to W and $1/\kappa_2 = 2$ years for the total decline from W to S): immunized individuals (R) migrate to the W compartment and then return to S, periodically replenishing the pool of susceptible individuals and allowing for new epidemic cycles. These oscillations are called endogenous and are the reason for one or more epidemic peaks. When seasonal forcing is introduced ($\varepsilon = 0.3$, Figure 3.3(b)), the endogenous cycles interact with the annual rhythm of $\beta(t)$. This interaction produces two observable effects: first, epidemic peaks are more synchronized with the seasons, tending to coincide with periods of high winter transmission; second, the amplitude of the peaks becomes higher and more variable from one cycle to the next, reflecting the combined effects of immune waning and seasonal fluctuations in the transmission rate. This result illustrates that seasonality creates epidemic cycles, modulates them, synchronizes them, and can amplify their intensity. These oscillations are called exogenous.

3.5.4 Deterministic SIRWS :Sensitivity to epsilon. We will observe how the seasonal amplitude modifies the oscillations. To do this, we will test four values of ε on four panels ($\varepsilon = 0.0, 0.10, 0.20, 0.35$)

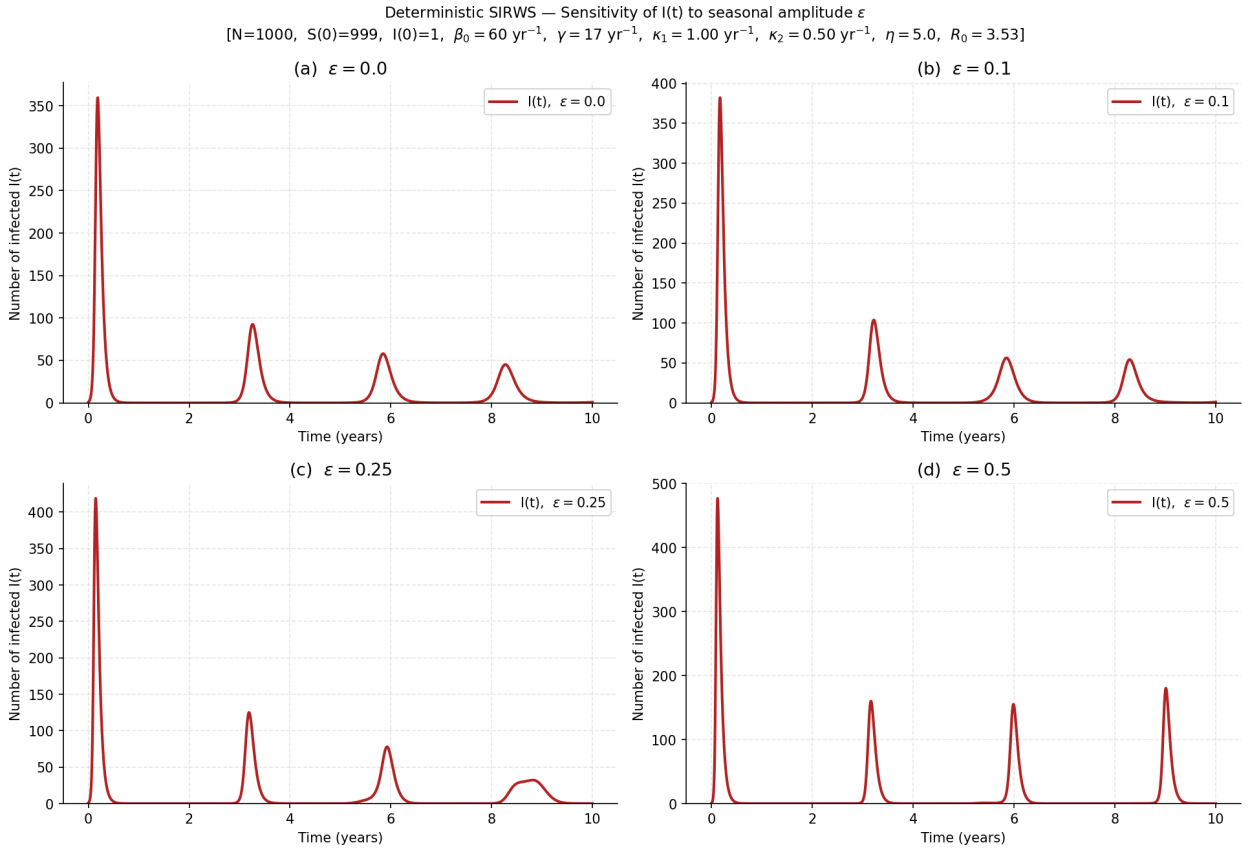


Figure 3.4: Dynamics over $T = 10$ years of the number of infected individuals in the SIRWS model and transmission peaks as a function of the seasonal forcing values, taking into account a duration of immunity of 3 years.

From Figure 3.4, we can observe that as seasonal forcing increases, the oscillations become more amplified and more regular.

4. Stochastic SIRWS and SIRS Models

The deterministic SIRWS model presented in the previous chapter describes the deterministic evolution of an epidemic in a large population. It assumes that individuals behave homogeneously and that the dynamics of the disease are entirely predictable once the initial conditions are fixed. In this chapter, we construct the stochastic version of the SIRWS model. We use the continuous-time counting process framework, which is the most rigorous and natural formulation for stochastic epidemic models. To capture these phenomena, we formulate the SIRWS model as a continuous-time Markov chain (CTMC), following the transition probability approach outlined by Allen [2] and Hridoy [10]. We also construct the corresponding stochastic SIRS model to compare the two dynamics.

In the Section 4.1, we will present the general framework of Continuous-Time Markov Chains (CTMC) and in the Section 4.2, we wil formulate the model.

4.1 Definition and theoretical framework of Continuous-Time Markov Chains

4.1.1 Definition. (Formal definition) [1].

A stochastic process $\{X(t)\}_{t\geq 0}$ is said to have a continuous-time Markov chain (CTMC) property if for all $0 \leq t_0 < t_1 < \cdots < t_n < t_{n+1}$ and for all state $i_0, i_1, \cdots, i_n, i_{n+1}$:

$$\Pr\{X(t_{n+1}) = i_{n+1} | X(t_n) = i_n, \cdots, X(t_0) = i_0\} = \Pr\{X(t_{n+1}) = i_{n+1} | X(t_n) = i_n\} \tag{4.1.1}$$

with $t \in [0, +\infty[$ and $i_k \in \{1, 2, 3, \cdots\}$ for $k = 0, 1, 2, \cdots, n$.

In simpler terms, the probability that the system is in state i_{n+1} at time t_{n+1} depends only on the present state i_n at time t_n, and not on the entire past history of the system. We say that the process is memoryless, or that it possesses the Markov property.

4.1.2 Infinitesimal transition probabilities. The local behavior of a CTMC is entirely characterized by its infinitesimal transition probabilities. Let us consider the CTMC given in Definition 4.1.1. For a small time interval Δt , the probability of transitioning from the state i to the state j is

$$p_{ij}(\Delta t) = \Pr\left(X(t + \Delta t) = j | X(t) = i\right).$$

More generally, using transition rates between states, the probability of transitioning from state x to state $x + h$ is given as follows

$$\Pr\left(X(t + \Delta t) = x + h | X(t) = x\right) = \psi(x, h)\Delta t + o(\Delta t), \tag{4.1.2}$$

where

- h is the transition vector that describes the change of state,
- $\psi(x, h) \geq 0$ is the transition rate associated with this change, and

- $o(\Delta t)$ is the Landau function with the properties $\lim_{h\to 0} \frac{o(h)}{h} = 0$, that is $o(h)$ goes faster to zero than h.

The probability that no event occurs in $[t, t + \Delta t)$ is given as

$$\Pr\left(X(t+\Delta t) = x | X(t) = x\right) = 1 - \sum_{j}^{M} \psi(x, h_j)\Delta t + o(\Delta t), \tag{4.1.3}$$

where M is the total number of all possible events since state x

4.2 Stochastic model formulation for SIRWS

Since $S+I+R+W = N$ is an exact constraint, then only three of the compartments are independent. Knowing three of them automatically determines the fourth. We keep S, I and W because these are the compartments with the richest epidemiological dynamics. We then choose the following state vector:

$$X(t) = (S(t), I(t), W(t)) \in \{0, 1, \cdots, N\}^3,$$

and we deduce at any time $t \geq 0$ the total recovered population is given as

$$R(t) = N - S(t) - I(t) - W(t).$$

We are primarily interested in epidemic dynamics, that is, how the disease spreads, generates cycles, dies out, or persists. For this reason, we will neglect demography, whose underlying mechanism is to maintain a constant population over the long term, without changing the structure of epidemic cycles. The SIRWS model without demography then includes 5 types of biological events. For each event, we provide the transition vector h_j, its biological description, and its infinitesimal probability. Recall that $R = N - S - I - W$ at any given time, so the rates are expressed entirely as functions of S, I and W. The model has five events the full description of which is given in Table 4.1.

Table 4.1: Table describing possible transitions and their rates, as well as biological descriptions

event	Transition: h_j	rate: $\psi(x, h_j)$	Description
1	$S \to S-1, I \to I+1$	$\beta(t)SI/N$	new infection
2	$I \to I-1, R \to R+1$	γI	new recover
3	$R \to R-1, W \to W+1$	$k_1 R$	partial loss of immunity
4	$W \to W-1, S \to S+1$	$k_2 W$	total loss of immunity
5	$W \to W-1, W \to R+1$	$\eta\beta(t)IW/N$	new boosting

Using Table 4.1, the transition probabilities are given as

$$\Pr\{\Delta X(t) = h|X(t)\} = \begin{cases} \frac{\beta(t)}{N} S(t)I(t)\Delta t + o(\Delta t) & if \quad h = (-1,+1,0), \\ \gamma I(t)\Delta t + o(\Delta t) & if \quad h = (0,-1,0), \\ \kappa_1(N - S(t) - I(t) - W(t))\Delta t + o(\Delta t) & if \quad h = (0,0,+1), \\ \kappa_2 W(t)\Delta t + o(\Delta t) & if \quad h = (+1,0,-1), \\ \eta\frac{\beta(t)}{N} I(t)W(t)\Delta t + o(\Delta t) & if \quad h = (0,0,-1), \\ 1 - \sum_j^5 \psi(X,h_j)\Delta t + o(\Delta t) & if \quad h = (0,0,0), \\ o(\Delta t) & otherwise \end{cases} \tag{4.2.1}$$

with $X = (S, I, W)$; $\Delta X(t) = X(t + \Delta t) - X(t)$ and

$$\sum_j^5 \psi(X,h_j)\Delta t = \left(\frac{\beta(t)}{N} SI + \gamma I + \kappa_1(N - S(t) - I(t) - W(t)) + \kappa_2 W + \eta\frac{\beta(t)}{N} IW\right)\Delta t,$$

The initial condition are given as $S(0) = s_0, I(0) = i_0$ and $W(0) = w_0$.

4.3 Kolmogorov equation for SIRWS model

There are two complementary forms of Kolmogorov's equations (forward and backward equations), each useful for different questions. In this paper, we use the forward equation, which answers the question: "Given the initial state, what is the probability of being in state x at time t ?" It describes the evolution of the probability distribution towards the future.

In this Section, we derive the forward Kolmogorov's equations for the SIRWS model. Let $p(x,t) = P(X(t) = x)$ denote the probability of being in state $x = (s,i,w)$ at time t. Using transition probabilities given in Equation (4.2.1), the Kolmogorov equation of the model system SIRWS is given by

$$\begin{aligned} \frac{dp_{(s,i,w)}}{dt} =& \frac{\beta(t)}{N}(s+1)(i-1) \times p_{(s+1,i-1,w)}(t) + \gamma(i+1)p_{(s,i+1,w)}(t) + \\ & \kappa_1(N - s - i - (w-1))p_{(s,i,w-1)}(t) + \kappa_2(w+1)p_{(s-1,i,w+1)}(t) + \\ & \eta\frac{\beta(t)}{N} i(w+1)p_{(s,i,w+1)}(t) + \\ & - \left(\frac{\beta(t)}{N} si + \gamma i + \kappa_1(N - s - i - w) + \kappa_2 w + \eta\frac{\beta(t)}{N} iw\right) p_{(s,i,w)}. \end{aligned} \tag{4.3.1}$$

For $s, i, w = 1, 2, 3, \cdots, N-1$. The boundary conditions are given as

$$\begin{aligned}
\frac{dp_{(0,0,0)}(t)}{dt} &= \gamma p_{(0,1,0)}(t) - \kappa_1 N p_{(0,0,0)}(t),\\
\frac{dp_{(N,0,0)}(t)}{dt} &= \kappa_2 p_{(N-1,0,1)}(t),\\
\frac{dp_{(0,N,0)}(t)}{dt} &= \frac{\beta(t)}{N}(N-1)p_{(1,N-1,0)}(t) - \gamma N p_{(0,N,0)}(t),\\
\frac{dp_{(0,0,N)}(t)}{dt} &= \gamma p_{(0,1,N)}(t) - \kappa_2 N p_{(0,0,N)},
\end{aligned}$$

and for $(0,0,w);(0,i,0);(s,0,0);(s,i,0);(s,0,w);(0,i,w)$, the equations are given as

$$\begin{aligned}
\frac{dp_{(s,0,0)}}{dt} &= \gamma p_{(s,1,0)}(t) + \kappa_2 p_{(s-1,0,1)}(t) - \kappa_1(N-s)p_{(s,0,0)}(t) \qquad s = 1,2\ldots N-1 \quad and \quad i = 0, w = 0\\
\frac{dP_{(0,i,0)}}{dt} &= \frac{\beta(t)}{N}(i-1)p_{(1,i-1,0)}(t) + \gamma(i+1)p_{(0,i+1,0)}(t) + \frac{\eta\beta(t)}{N} i p_{(0,i,1)}(t) - \big[\gamma i + \kappa_1(N-i)\big]p_{(0,i,0)}(t)\\
&\quad i = 1,2\ldots N-1 \quad and \quad s = 0, w = 0,\\
\frac{dP_{(0,0,w)}}{dt} &= \gamma p_{(0,1,w)}(t) + \kappa_1(N-w+1)p_{(0,0,w-1)}(t) - \big[\kappa_1(N-w) + \kappa_2 w\big]p_{(0,0,w)}(t),\\
\frac{dp_{(s,i,0)}}{dt} &= \frac{\beta(t)}{N}(s+1)(i-1)\,P_{(s+1,i-1,0)}(t) + \gamma(i+1)p_{(s,i+1,0)}(t) + \kappa_2 p_{(s-1,i,1)}(t)\\
&\quad + \frac{\eta\beta(t)}{N} i p_{(s,i,1)}(t) - \Big[\frac{\beta(t)}{N} si + \gamma i + \kappa_1(N-s-i)\Big]p_{(s,i,0)}(t) \qquad w = 0 \quad and \quad s,i = 1,2\ldots N-1,\\
\frac{dp_{s,0,w}}{dt} &= \gamma p_{(s,1,w)}(t) + \kappa_1(N-s-w+1)p_{s,0,w-1}(t) + \kappa_2(w+1)p_{s-1,0,w+1}(t)\\
&\quad - \big[\kappa_1(N-s-w) + \kappa_2 w\big]p_{(s,0,w)}(t) \qquad i = 0 \quad and \quad s,w = 1,2\ldots N-1,\\
\frac{dp_{(0,i,w)}}{dt} &= \frac{\beta(t)}{N}(i-1)p_{(1,i-1,w)}(t) + \gamma(i+1)p_{(0,i+1,w)}(t) + \kappa_1(N-i-w+1)p_{(0,i,w-1)}(t) +\\
&\quad \frac{\eta\beta(t)}{N} i(w+1)p_{(0,i,w+1)}(t) - \Big[\gamma i + \kappa_1(N-i-w) + \kappa_2 w + \frac{\eta\beta(t)}{N} iw\Big]p_{(0,i,w)}(t) \qquad s = 0\\
&\quad and \quad i,w = 1,2\ldots N-1.
\end{aligned}$$

4.4 Stochastic model formulation for SIRS

The SIRS model is the simplified version of the SIRWS model without W compartment or a boosting mechanism. The structure is $S \to I \to R \to S$, with a direct loss of immunity from R to S at the rate κ. We can observe the flow diagram in Figure (4.1).

The system state is given as

$$Y(t) = (S(t), I(t)) \in \{0, 1, \cdots, N\}^2,$$

and we deduce at any time the recovered population is given as

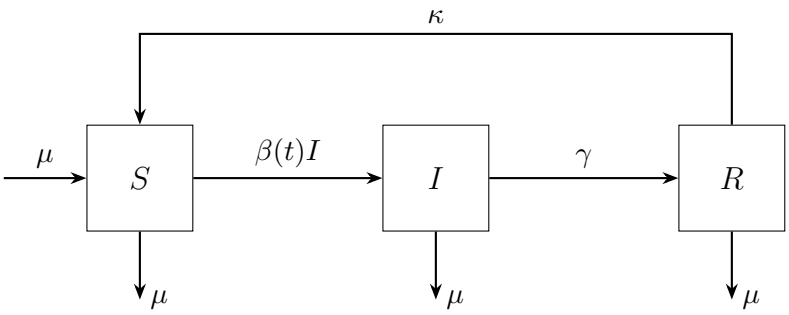


Figure 4.1: Flow diagram of the model

$$R(t) = N - S(t) - I(t).$$

The model has three events, the full description of which is given in Table 4.2.

Table 4.2: Table describing possible transitions and their rates, as well as biological descriptions

events rank	Transitions: h_j	rates: $q_j(x, h_j)$	Descriptions
1	$S \to S-1, I \to I+1$	$\beta(t)SI/N$	new infection
2	$I \to I-1, R \to R+1$	γI	new recovery
3	$R \to R-1, S \to S+1$	κR	waning

Using Table 4.2, the transition probabilities are given as

$$Pr\{\Delta Y(t) = h/Y(t)\} = \begin{cases} \frac{\beta(t)}{N} S(t)I(t)\Delta t + o(\Delta t) & if \quad h = (-1,+1), \\ \gamma I(t)\Delta t + o(\Delta t) & if \quad h = (0,-1), \\ \kappa(N - S(t) - I(t))\Delta t + o(\Delta t) & if \quad h = (1,0), \\ 1 - \left(\frac{\beta(t)}{N} S(t)I(t) + \gamma I(t) + \kappa(N - S(t) - I(t))\right)\Delta t + o(\Delta t) & if \quad h = (0,0) \\ o(\Delta t) & otherwise \end{cases} \tag{4.4.1}$$

with $\Delta Y(t) = Y(t + \Delta t) - Y(t)$ and initial condition $S(0) = s_0, I(0) = i_0$.

Here, reasoning in the same way as in Section (4.3), we derived the forward Kolmogorov equation for SIRS Model. Let $p(y,t) = P(Y(t) = y)$ be the probability of being in the state $y = (s,i)$ at time t. Then, the Kolmogorov forward equation is expressed mathematically as

$$\begin{aligned} \frac{dp_{(s,i)}(t)}{dt} &= \beta(t)\frac{(s+1)(i-1)}{N} p_{(s+1,i-1)}(t) + \gamma(i+1)\, p_{(s,i+1)}(t) + \kappa(N-i-s+1)\, p_{(s-1,i)}(t) \\ &\quad - \left(\beta(t)\,\frac{si}{N} + \gamma i + \kappa(N-s-i)\right) p_{(s,i)}(t), \qquad \text{if } s,i > 0 \text{ and } s+i < N, \end{aligned} \tag{4.4.2}$$

with the boundary conditions given as

$$\begin{aligned}
\frac{dp_{(s,0)}(t)}{dt} &= \gamma p_{(s,1)}(t) + \kappa(N-s+1)p_{(s-1,0)}(t) - \kappa(N-s)p_{(s,0)}(t), \qquad s = 1, 2, \ldots, N-1, \\
\frac{dp_{(0,i)}(t)}{dt} &= \beta(t)\frac{(i-1)}{N}p_{(1,i-1)}(t) + \gamma(i+1)p_{(0,i+1)}(t) - \big(\gamma i + \kappa(N-i)\big)p_{(0,i)}(t), \qquad i = 1, 2, \ldots, N-1, \\
\frac{dp_{0,0}(t)}{dt} &= \gamma p_{(0,1)}(t) - \kappa N p_{(0,0)}(t) \qquad s = 0, i = 0, \\
\frac{dp_{(N,0)}(t)}{dt} &= \gamma p_{(N-1,1)}(t) + \kappa p_{(N-1,0)}(t) \qquad s = N, i = 0, \\
\frac{dp_{0,N}(t)}{dt} &= \beta(t)\,\frac{N-1}{N}p_{(1,\,N-1)}(t) - \gamma N p_{(0,N)}(t) \qquad s = 0, i = N.
\end{aligned}$$

4.5 Expected value, variance and extinction probability

4.5.1 Approximation close to the DFE. As mentioned in the section on the deterministic model in Chapter 3, near the disease-free equilibrium (DFE), the number of infectious individuals in the SIRS and SIRWS models can be approximated by the linear differential equation:

$$\frac{dI(t)}{dt} \approx \big(\beta(t) - \gamma_e\big)\, I(t), \tag{4.5.1}$$

where the natural mortality rate μ is absorbed into γ_e to simplify the notation. This approximation is valid as long as $S \approx N$, that is, at the beginning of an epidemic or in the vicinity of the DFE.

The equivalent stochastic process for the seasonally forced SIRS and SIRWS models is a time-inhomogeneous birth-death process, whose forward Kolmogorov equation is written:

$$\frac{dp_i(t)}{dt} = \begin{cases} \beta(t)(i-1)p_{i-1}(t) - \big(\beta(t) + \gamma_e\big)ip_i(t) + \gamma_e(i+1)p_{i+1}(t), & i = 1, 2, \ldots \\ \\ \gamma_e p_1(t), & i = 0, \end{cases} \tag{4.5.2}$$

with the initial condition $p_{i_0}(t_0) = 1$ where $I(0) = i_0$, under the assumption that N is sufficiently large. If N is finite, the boundary condition must be included:

$$\frac{dp_N(t)}{dt} = \beta(t)(N-1)p_{N-1}(t) - \gamma_e N p_N(t).$$

Because when $i = N$, the entire population is infectious. There are no more susceptible individuals available in the population. The infection response is no longer possible. Hence, $\beta(t) = 0$ for $i = N$

4.5.2 Expectation and variance of $I(t)$. To analytically analyze the expectation and variance of $I(t)$, we use the cumulating-generating function, defined by:

$$\mathcal{K}(\theta, t) = \log \Phi(e^{\theta}, t), \tag{4.5.3}$$

where $\Phi(z, t) = \mathbb{E}[z^{I(t)}]$ is the probability-generating function (PGF).

In the general case, it can be shown that the probability generating function can be written as:

$$\Phi(z,t) = \left(1 + \frac{1}{\dfrac{e^{\rho(t)}}{z-1} + L(t)}\right)^{i_0}, \tag{4.5.4}$$

where the two key quantities are defined as

$$\begin{aligned}\rho(t) &= \gamma_e t - \int_0^t \beta(s)\mathrm{d}s, \\ &= \gamma_e t - \beta_0 \int_0^t \left(1 + \varepsilon\cos\left(\frac{2\pi s}{\omega} + \phi\right)\right) ds, \\ &= (\gamma_e - \beta_0)t - \frac{\beta_0 \varepsilon \omega}{\pi}\left[\cos\left(\frac{\pi t}{\omega} + \phi\right)\sin\left(\frac{\pi t}{\omega}\right)\right],\end{aligned} \tag{4.5.5}$$

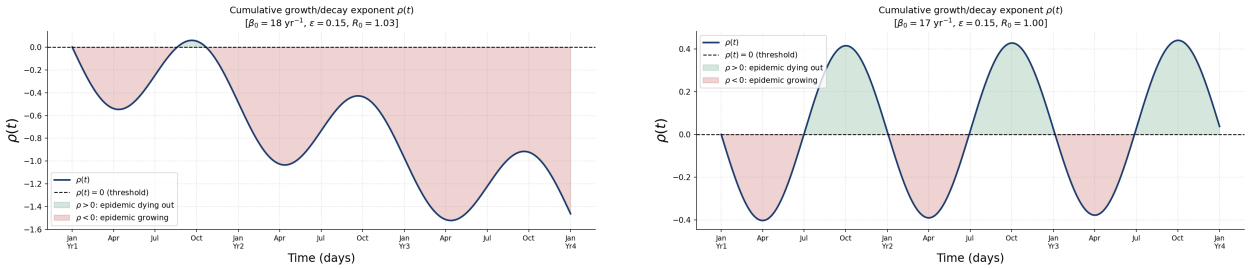
is the cumulative exponent of the growth or decay of the process, and

$$L(t) = \int_0^t \beta(s)\, e^{\rho(s)}\mathrm{d}s \tag{4.5.6}$$

is the time-weighted integral of transmission rate.

4.5.3 Remark. If $\rho(t) \to +\infty$ when $t \to +\infty$, the process die out almost surely, and if $\rho(t) \to -\infty$, the process persists and the probability of extinction is $p_0(t) = \Phi(0,t)$.

We can observe it in Figure (4.2).



(a) $\rho(t)$ curve for $R_0 > 1$ ($\beta_0 = 18 > \gamma_e$)

(b) $\rho(t)$ curve for $R_0 < 1$ ($\beta_0 = 17 < \gamma_e$)

Figure 4.2: Cumulative exponent of the growth or decay of the process for $R_0 > 1$ (the epidemic spread very effectively) and $R_0 < 1$ (the epidemic gradually died out).simulation over T=3 years with $I_0 = 1$, $\varepsilon = 0.15$ The oscillations of $\rho(t)$ reflect the seasonal variations of $\beta(t)$

4.5.4 Proposition. The expectation of $I(t)$ in the stochastic process is given by

$$\mathbb{E}\big[I(t)\big] = I(0)\, e^{-\rho(t)}. \tag{4.5.7}$$

Proof. The expected value is given by $\mathbb{E}[I(t)] = \Phi'(1,t)$, where $\Phi'(1,t)$ is the derivative of $\Phi(z,t)$ with respect to z and evaluated at $z = 1$. By calculating the derivative, we obtain the result above. □

4.5.5 Remark. The expectation in Equation (4.5.7) coincides exactly with the solution of the linearized deterministic model (3.4.8). The expectation of the stochastic process therefore equals the deterministic solution.

4.5.6 Proposition. The variance of $I(t)$ in the stochastic process is given by

$$\mathrm{Var}\big[I(t)\big] = I(0)\, e^{-\rho(t)} \int_0^t \big(\beta(s) + \gamma_e\big)\, e^{\rho(s)}\, \mathrm{d}s. \tag{4.5.8}$$

The variance increases with time as long as $\beta(s) + \gamma_e > 0$

Proof. The expected value is given by $\mathbb{V}[I(t)] = \mathbb{E}[I^2(t)] - \mathbb{E}[I(t)]^2 = \mathbb{E}[I(t)(I(t)-1)] + \mathbb{E}[I(t)] - \mathbb{E}[I(t)]^2$. But $\mathbb{E}[I(t)(I(t)-1)] = \Phi''(1,t)$, where $\Phi''(1,t)$ is the second derivative of $\Phi(z,t)$ with respect to z and evaluated at $z = 1$. By calculating of the variance, we obtain the result above. □

Special case: if $\beta(t)$ was not time-dependent, that is $\beta(t) = \beta_0$ (constant) with $\beta_0 \neq \gamma_e$, then:

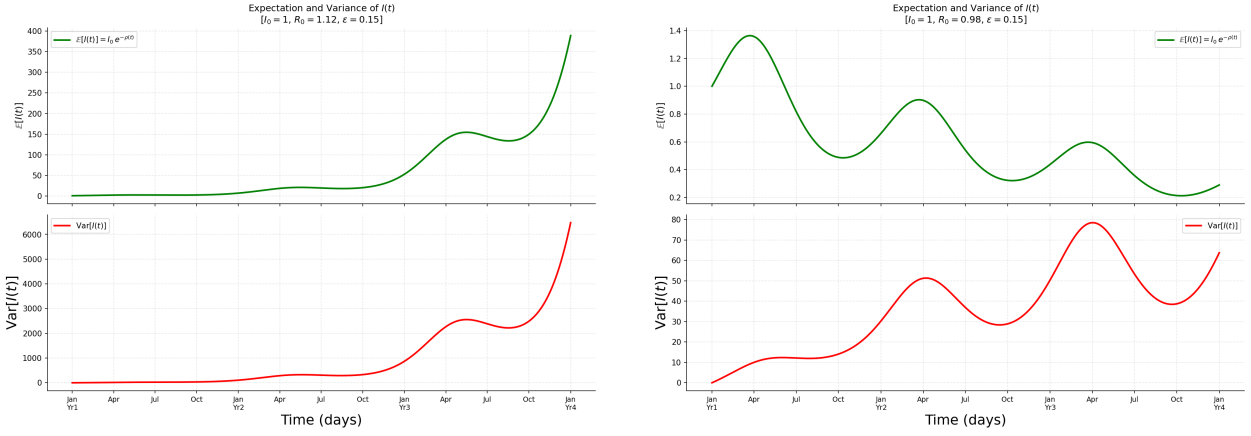
$$\mathbb{E}\big[I(t)\big] = I(0)\, e^{(\beta_0 - \gamma_e)t} = I(0)\, e^{(\gamma+\mu)(R_0-1)t}, \tag{4.5.9}$$

$$\mathrm{Var}\big[I(t)\big] = I(0) \left(\frac{\beta_0 + \gamma_e}{\beta_0 - \gamma_e}\right) e^{(\beta_0-\gamma_e)t} \Big(e^{(\beta_0-\gamma_e)t} - 1\Big). \tag{4.5.10}$$

At the epidemic threshold $\beta_0 = \gamma_e$ (taking the limit):

$$\mathrm{Var}\big[I(t)\big] = I(0) \cdot 2\beta_0\, t. \tag{4.5.11}$$

The variance grows linearly at the threshold.



(a) The expected value and variance of $I(t)$ ($\beta_0 = 18 > \gamma_e$)

(b) The expected value and variance of $I(t)$ ($\beta_0 = 17 < \gamma_e$)

Figure 4.3: Expected value and variance curve of I(t) for $R_0 < 1$ and $R_0 > 1$, simulated over T=3 years with $I_0 = 1$, $\varepsilon = 0.15$ and $\phi = 0$ radian

In Figure 4.3(a) the expectation of $I(t)$ grows exponentially with time, starting from $I_0 = 1$. Since $R_0 > 1$, $\rho(t) \to -\infty$ and therefore $e^{-\rho(t)} \to +\infty$. The variance follows the same pattern, but grows

even faster than the expected value (its variations can be observed). In Figure 4.3(b), we observe the opposite phenomenon, since here $R_0 < 1$, $\rho(t) \to +\infty$ and therefore $e^{-\rho(t)} \to 0$.

The curve exhibits undulations due to the seasonality of $\beta(t)$. In winter, growth accelerates. In summer it slows slightly, which explains the resurgence of new waves of the epidemic. The expected value coincides exactly with the deterministic solution. The variance grows even faster than the expected value. This means that the further the epidemic progresses, the greater the dispersion of the stochastic trajectories. In other words, two epidemics that start with exactly one infectious agent at the same time can unfold very differently. This is the signature of stochasticity that the deterministic model cannot capture.

4.5.7 Probality of extinction. The probability of extinction at time t, denoted $p_0(t)$, is the probability that $I(t) = 0$. It is obtained from the probability generating function evaluated at $z = 0$. We then have:

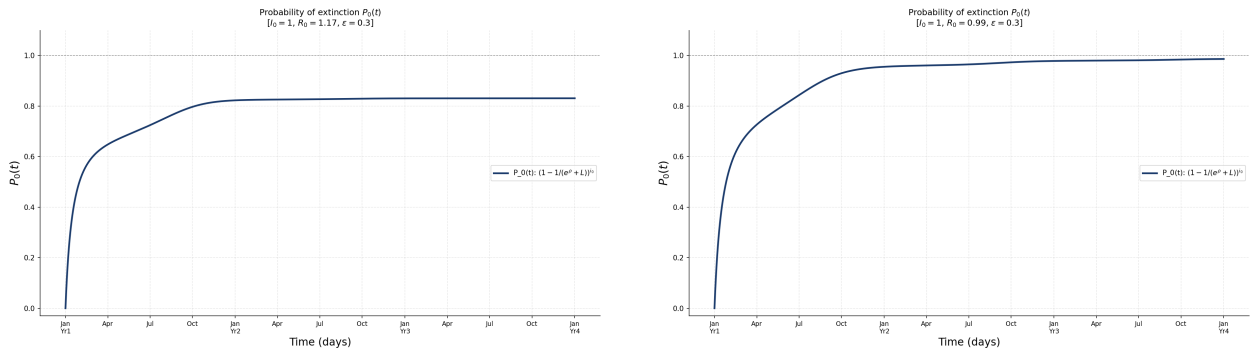
$$p_0(t) = \Phi(0,t) = \left(1 - \frac{1}{e^{\rho(t)} + L(t)}\right)^{i_0}. \tag{4.5.12}$$

Expanding explicitly using the definition of $L(t)$:

$$p_0(t) = \left(\frac{\gamma_e \int_0^t e^{\rho(s)}\,\mathrm{d}s}{1 + \gamma_e \int_0^t e^{\rho(s)}\,\mathrm{d}s}\right)^{i_0}. \tag{4.5.13}$$

Asymptotic behavior: It is evident that:

$$\lim_{t\to\infty} p_0(t) = 1 \quad \text{if and only if} \quad \lim_{t\to\infty} \rho(t) = +\infty. \tag{4.5.14}$$



(a) $\rho(t)$ curve for $R_0 > 1$ ($\beta_0 = 18 > \gamma_e$) (b) $\rho(t)$ curve for $R_0 < 1$ ($\beta_0 = 17 < \gamma_e$)

Figure 4.4: Probability curve of epidemic extinction for $R_0 < 1$ and $R_0 > 1$, simulated over T=3 years with $I_0 = 1$, $\varepsilon = 0.3$ and $\phi = 0$ radian

The curves in Figure 4.4 confirms the previous results. In Figure 4.4(a), we clearly observe that the probability of extinction is far from the neighborhood of 1, indicating that the disease does not die easily. This explains the exponential growth of the infection in the previous figure. In Figure 4.4(b), we clearly see a rapid growth of P_0 in the vicinity of 1 just after the first year. This explains why the disease disappears.

4.6 Gillespie's Algorithm

Gillespie's algorithm [9] is the direct mathematical consequence of infinitesimal transition probabilities. It generates exact trajectories of the CTMC without any approximation of the time step.

Gillespie's Algorithm

Entrances : $\beta_0, \varepsilon, \varphi, \gamma, \kappa_1, \kappa_2, \eta, N, S(0), I(0), R(0), W(0), T$

Outputs: Trajectory $(S(t), I(t), R(t), W(t))$

1. $\mathbf{T}[1] \leftarrow 0$
2. $\mathbf{Q}[1] \leftarrow (S(0), I(0), R(0), W(0))$
3. $i \leftarrow 2$
4. $\beta \leftarrow \beta_0(1 + \varepsilon \cos(2\pi \mathbf{T}[i-1]/\omega + \varphi))$
5. $\psi_1 = \frac{\beta}{N} IS,\ \psi_2 = \gamma I,\ \psi_3 = \kappa_1 R,\ \psi_4 = \kappa_2 W,\ \psi_5 = \frac{\eta\beta}{N} IW$
6. $\psi^* \leftarrow \sum_{j=1}^{5} \psi_j$
7. $\Delta t \leftarrow -\ln(U)/\psi^*$ with $U \sim \mathcal{U}(0,1)$
8. $\mathbf{T}[i] \leftarrow \mathbf{T}[i-1] + \Delta t$
9. $q_j \leftarrow \psi_j/\psi^*$ for $j = 1, \ldots, 5$
10. $\alpha \leftarrow \mathcal{U}(0,1)$
11. Find j such as $\sum_{k=1}^{j-1} q_k \leq \alpha < \sum_{k=1}^{j} q_k$
12. Update according to j :

- $j = 1 :\ S \leftarrow S - 1,\ I \leftarrow I + 1$
- $j = 2 :\ I \leftarrow I - 1,\ R \leftarrow R + 1$
- $j = 3 :\ R \leftarrow R - 1,\ W \leftarrow W + 1$
- $j = 4 :\ W \leftarrow W - 1,\ S \leftarrow S + 1$
- $j = 5 :\ W \leftarrow W - 1,\ R \leftarrow R + 1$

13. $\mathbf{Q}[i] \leftarrow (S, I, R, W)$
14. If $\mathbf{T}[i] \geq T$ or $I = 0$: STOP, return $\mathbf{T}, \mathbf{Q}$

Otherwise: $i \leftarrow i + 1$ and return to step 4

4.7 Simulations using Gillespie algorithm

Stochastic simulations are performed in Python using Gillespie's algorithm given above in Section 4.6, which generates exact trajectories of the continuous-time Markov chain (CTMC) without approximating the time step. At each iteration, the algorithm pulls the time to the next event from an exponential distribution with parameter $q_{tot}(t, \mathbf{X}(t)) = \sum_{j=1}^{5} q_j(t, \mathbf{X}(t))$, and selects the event that occurs

with a probability proportional to its rate. Since $\beta(t)$ depends explicitly on time, the CTMC is time-inhomogeneous; at each iteration, $\beta(t)$ is recalculated at the current time t before the transition rates are calculated. For the parameter values, refer again to Table 3.3 given in Chapter 3.

4.7.1 Stochastic SIRWS vs ODE: Observation over ONE year. Figure 4.5 illustrates the random nature of individual events (an infection, a recovery, a waning). The extent of the variability will be clearly visible.

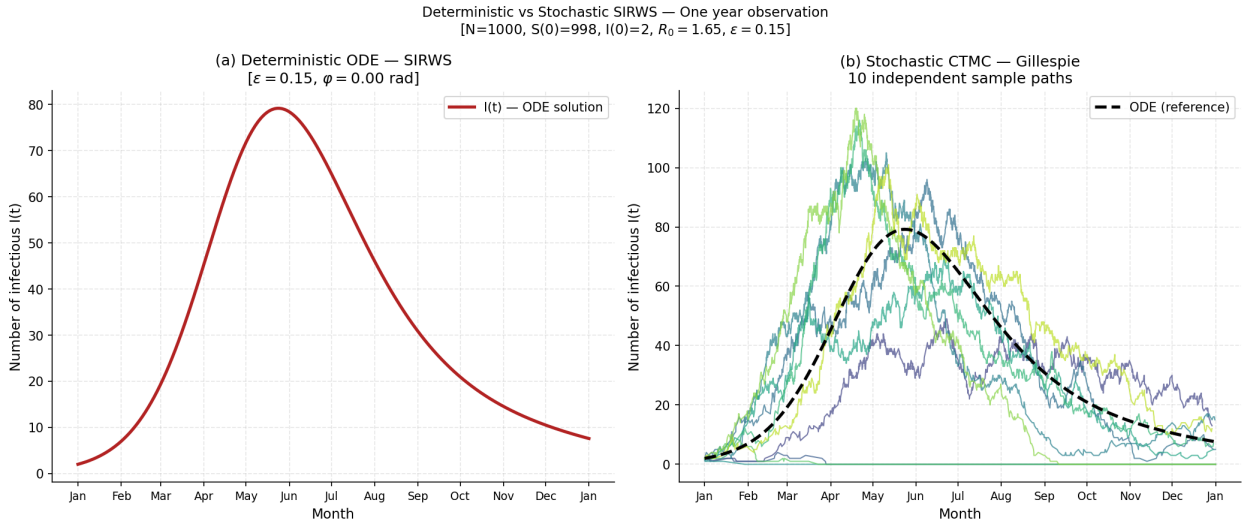


Figure 4.5: Comparison over $T = 1$ year between the deterministic solution and ten stochastic trajectories of the SIRWS model using the Gillespie algorithm, illustrating the stochastic variations with $\beta_0 = 30\text{yr}^{-1}$.

Panel (a) in Figure 4.5 shows the deterministic solution: a smooth, predictable, single curve. It represents the average behavior of a population. It answers the question: "On average, how many infected individuals will there be?" It cannot account for the random nature of individual events (an infection, a recovery, a waning). Panel (b) in Figure 4.5 shows 10 stochastic trajectories derived from the same model with the same initial conditions. Three major differences are observed.

The first observation is that there is significant variability between the trajectories. Each stochastic trajectory is unique. Some have a high peak, others a lower peak (< 60 infected individuals). Some reach their peak earlier, others later. This variability reflects the random nature of transmission stipulating that, by chance, an infected individual may quickly infect several people or, conversely, recover without infecting anyone.

The second observation is that the stochastic extinction when $R_0 > 1$. We observe that some trajectories die out ($I(t) = 0$) before the end of the year. However, $R_0 \approx 1,65 > 1$, so the deterministic model predicts that the epidemic persists. Stochasticity allows for extinction by chance, even when $R_0 > 1$. This is impossible in the deterministic model.

Moreover, the third observation is the stochastic phase shift. The peak of the ODE occurs at a precise time (determined by ϕ). Stochastic peaks, however, are staggered in time: some arrive earlier, others later. This is called stochastic phase shift. It explains why real epidemics do not always peak on exactly the same date each year.

These phenomena, impossible to capture by a deterministic model, justify the use of a stochastic model to represent epidemiological reality.

4.7.2 Stochastic SIRWS vs SIRS: observation over ten years. This presentation compares the stochastic dynamics of SIRWS and SIRS

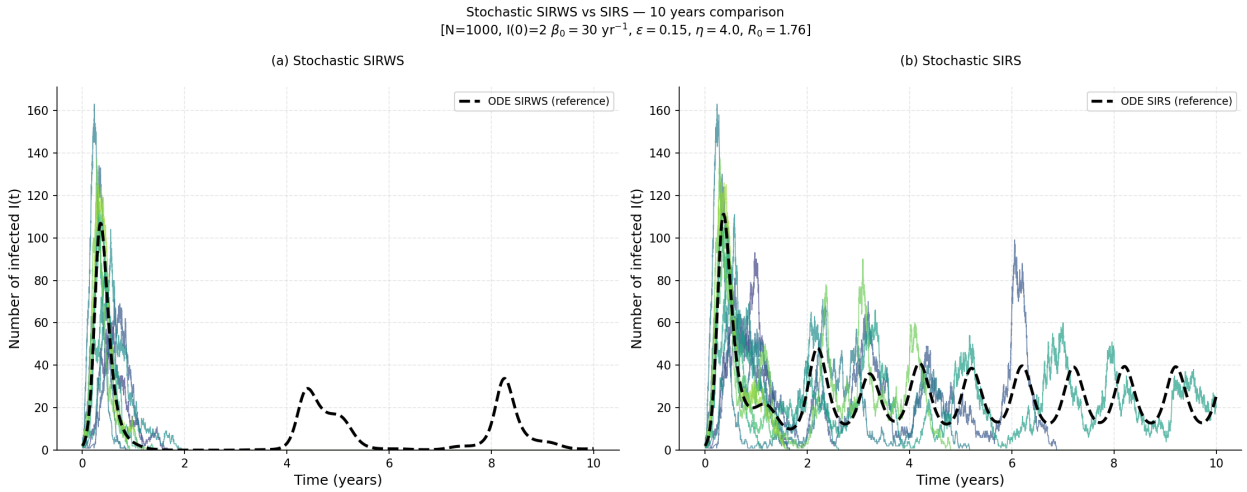


Figure 4.6: Comparison over 10 years between the dynamics of the stochastic models SIRWS and SIRS with $\varepsilon = 0.15$.

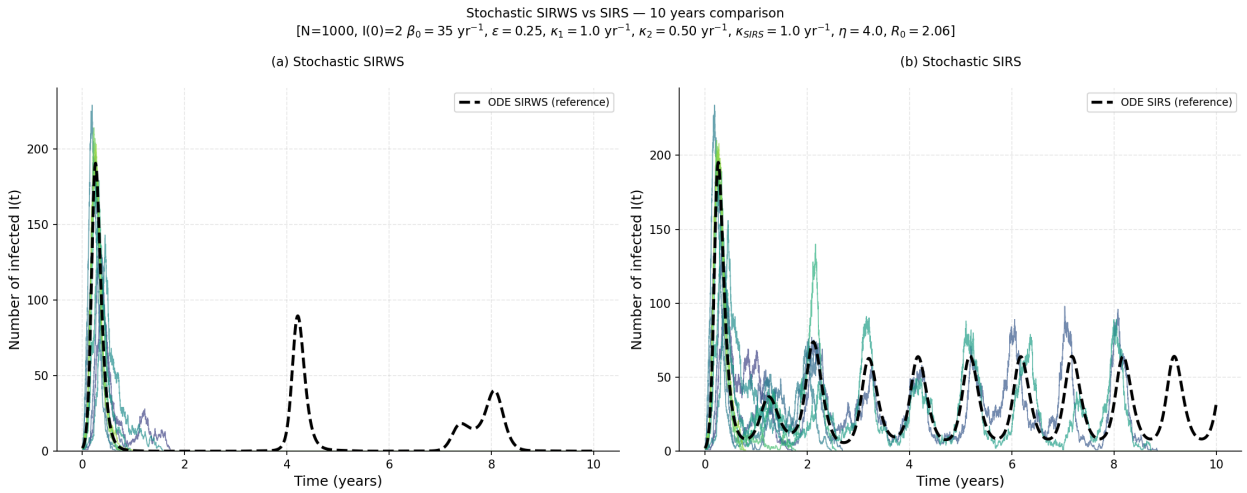


Figure 4.7: Comparison over 10 years between the dynamics of the stochastic models SIRWS and SIRS with $\varepsilon = 0.25$.

As can be seen in Figure 4.6-4.7, initially, at the first peak, the rapid spread of the epidemic dominates random behavior, causing the trajectories to appear convergent. Subsequently, once the epidemic stabilizes in an endemic phase, randomness begins to play an increasing role. Small reinfections cause the trajectories to diverge. In Figure 4.7 ($\varepsilon = 0.25$ and peak almost 250), the oscillations are more amplified than in Figure 4.6 ($\varepsilon = 0.15$), due to the amplification of seasonality. The presence of the W compartment, which models the loss of immunity in the SIRWS model, introduces a prolonged

extinction phase of the disease before its resurgence. This phase is not observed in the SIRS model, where the oscillations are more synchronized and the epidemic never reaches extinction (epidemic waves occur with very regular periods). It is always this W compartment that is responsible for the premature extinction of all stochastic trajectories in the SIRWS model, because recycling occurs slowly and the peak is no longer sufficient to maintain the trajectories. In the standard SIRS model, stochastic trajectories oscillate around the deterministic solution with inter-trajectory variability until the end of the 10-year period. Although some trajectories become extinct as the dynamics progress, this is typical of stochastic models.

4.7.3 The effect of immune boosting: an original contribution. This Figure 4.8 shows the effect of the boosting parameter η on the SIRWS stochastic model, for four increasing values: $\eta = 0$ (no boosting), $\eta = 1$ (weak), $\eta = 3$ (moderate), and $\eta = 5$ (strong). Each panel shows 10 independent stochastic trajectories (various colors) superimposed on the corresponding deterministic solution (black dashed curve).

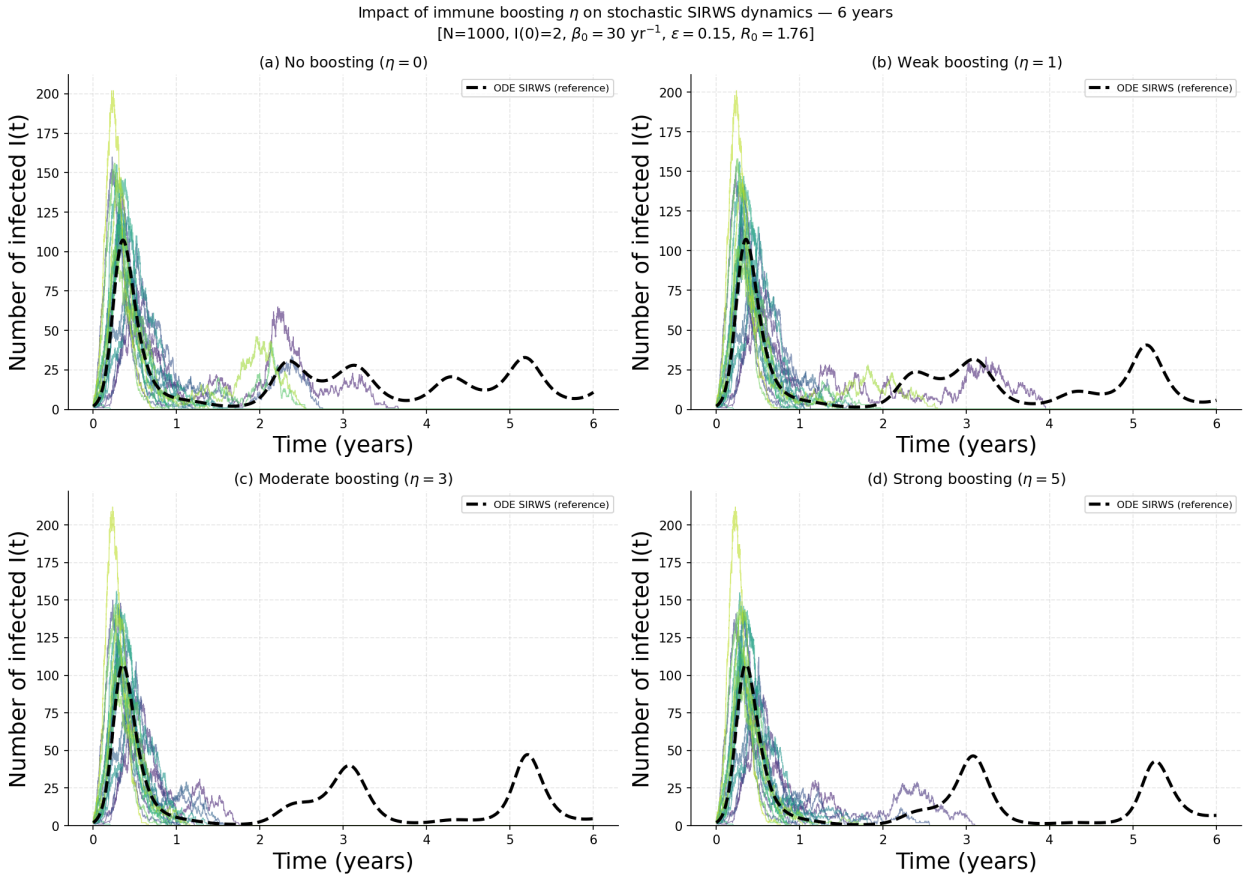


Figure 4.8: Observation over 6 years of the boosting effect on the stochastic dynamics of the SIRWS model.

For $\eta = 0$, there is no return to complete immunity. The system behaves like a standard SIRS model. Individuals simply lose their immunity and quickly become vulnerable, allowing the disease to spread rapidly and for a period without extinction. The dynamics follow the deterministic mean trend. For a low rate of return to complete immunity ($\eta = 1$), a slight delay is observed in the emergence of a new epidemic wave. Inter-trajectory variability begins to dampen. For $\eta = 3$, enhanced immunity protects

the population for a longer period. A prolonged extinction period of the epidemic is observed, which spaces out the peaks between successive epidemics and tends to attenuate the system's oscillations and lead to a faster extinction of most trajectories. For $\eta = 5$, a saturation effect is observed. The high rate of immunity enhancement constantly protects the majority of the population, thus preventing the emergence of a new major epidemic peak. This phenomenon reduces the effect of stochastic randomness. The epidemic then becomes endemic, characterized by a low and constant presence, without a significant peak.

5. Conclusion

This thesis aimed to analyze how seasonal forcing of the transmission rate affects the periodicity of epidemics in a SIRWS model incorporating progressive loss of immunity (waning) and immune restoration through re-exposure (boosting), by extending the analysis from the deterministic framework to the stochastic framework based on continuous-time Markov chains (CTMC). The approach followed a logical three-step progression: first, the analysis of the deterministic model with constant and then seasonal transmission rates; second, the rigorous construction of the stochastic model using infinitesimal transition probabilities and Kolmogorov equations; and finally, the comparison of the two frameworks through numerical simulation using the Gillespie algorithm.This approach yielded analytical and numerical results that mutually illuminate the two modeling frameworks.

It is essential to emphasize that the deterministic and stochastic frameworks are mathematically fundamentally different. The main analytical result is the demonstration that the extinction condition in the stochastic model is completely different from the deterministic model. In the deterministic framework, R_0 is a constant that alone determines whether the disease dies out or becomes endemic. In the stochastic framework, what matters is the cumulative exponent $\rho(t)$, which depends on time. The extinction condition is not exactly "$R_0 < 1$" but "$\rho(t) \to +\infty$". Another result observed during the analysis is that the deterministic model produces a unique, smooth, and perfectly predictable trajectory for fixed initial conditions: it is a constant (zero variance). The stochastic model, on the other hand, defines a probability distribution over the space of all possible trajectories: The number of infected $I(t)$ is a random variable, and each realization is unique. The expected value and variance are well-defined quantities of this distribution. The variance, however, grows faster than the expected value, revealing that epidemic uncertainty is a fundamental property of the stochastic process that the deterministic model alone cannot capture.

Numerical simulations using the Gillespie algorithm revealed three fundamental phenomena absent from the deterministic framework: stochastic extinction (an epidemic can die out by chance even when $R_0 > 1$), stochastic phase shift (epidemic peaks do not occur at exactly the same time as predicted by the ODE), and inter-trajectory variability (two epidemics with identical initial conditions can produce very different dynamics). These phenomena, observed in real epidemiological data but inexplicable by deterministic models, find a natural justification within the stochastic framework. The comparison of stochastic SIRWS and SIRS models constitutes the most significant original contribution of this thesis. It shows, for the first time within a stochastic framework, that the immune boosting mechanism acts as a stabilizer of epidemic dynamics: by recycling a fraction of individuals with declining immunity (W) to full immunity (R) without them contracting the disease, boosting maintains a more stable level of herd immunity, lengthens inter-epidemic cycles, and reduces the stochastic variability of trajectories. This finding, absent from previous literature which addressed boosting solely within deterministic frameworks [5, 6, 12], opens new perspectives for understanding the dynamics of diseases with non-permanent immunity, such as pertussis or influenza.

More broadly, this work illustrates that seasonality and stochasticity are distinct mechanisms. Seasonality is a deterministic forcing: $\beta(t)$ is a known and predictable function of time that synchronizes and amplifies epidemic cycles. Stochasticity is irreducible randomness: it introduces uncertainty about the timing and magnitude of epidemics that no deterministic function can capture. Simultaneously considering these two mechanisms in a model integrating weakening and boosting is essential for accurately representing the complexity of real-world epidemics.

Several natural extensions to this work are possible. Incorporating an age structure would allow for a more

accurate modeling of pertussis, whose dynamics are strongly influenced by school contacts. Calibrating the parameters using real-world epidemiological surveillance data would give the results predictive value. Studying the epidemic probability $P_{\text{outbreak}}(t_0)$ as a function of the season of introduction, within the framework of the complete SIRWS model (and not simply the linearized birth-death process), would constitute a rich analytical extension. Finally, analyzing the effect of vaccination campaigns on the stochastic dynamics of the SIRWS model, particularly their interaction with the natural boosting mechanism, represents a highly relevant public health issue given the observed resurgence of pertussis in several countries with high vaccination coverage.

This thesis thus hopes to make a modest but significant contribution to the mathematical understanding of the periodicity of infectious diseases, by demonstrating that the dynamic complexity of these epidemics can only be fully grasped by simultaneously considering seasonality, randomness, and the complex mechanisms of immunity.

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References

[1] Linda JS Allen. *An introduction to stochastic processes with applications to biology*. CRC press, 2010.

[2] Linda JS Allen. Stochastic population and epidemic models. *Mathematical biosciences lecture series, stochastics in biological systems*, 1:120–128, 2015.

[3] Roy M Anderson and Robert M May. *Infectious diseases of humans: dynamics and control*. Oxford university press, 1991.

[4] Emmanuelle Augeraud-Véron and Nadir Sari. Seasonal dynamics in an sir epidemic system. *Journal of mathematical biology*, 68(3):701–725, 2014.

[5] Mathew P Dafilis, Federico Frascoli, Jodie McVernon, Jane M Heffernan, and James M McCaw. The dynamical consequences of seasonal forcing, immune boosting and demographic change in a model of disease transmission. *Journal of Theoretical Biology*, 361:124–132, 2014.

[6] Mathew P Dafilis, Federico Frascoli, James G Wood, and James M McCaw. The influence of increasing life expectancy on the dynamics of sirs systems with immune boosting. *The ANZIAM Journal*, 54(1-2):50–63, 2012.

[7] Klaus Dietz. The estimation of the basic reproduction number for infectious diseases. *Statistical methods in medical research*, 2(1):23–41, 1993.

[8] David JD Earn, Pejman Rohani, Benjamin M Bolker, and Bryan T Grenfell. A simple model for complex dynamical transitions in epidemics. *science*, 287(5453):667–670, 2000.

[9] Daniel T Gillespie. Exact stochastic simulation of coupled chemical reactions. *The journal of physical chemistry*, 81(25):2340–2361, 1977.

[10] Mahmudul Bari Hridoy. An exploration of modeling approaches for capturing seasonal transmission in stochastic epidemic models. *arXiv preprint arXiv:2410.16664*, 2024.

[11] William Ogilvy Kermack and Anderson G McKendrick. A contribution to the mathematical theory of epidemics. *Proceedings of the royal society of london. Series A, Containing papers of a mathematical and physical character*, 115(772):700–721, 1927.

[12] Jennie S Lavine, Aaron A King, and Ottar N Bjørnstad. Natural immune boosting in pertussis dynamics and the potential for long-term vaccine failure. *Proceedings of the National Academy of Sciences*, 108(17):7259–7264, 2011.

[13] Tiffany Leung, Barry D Hughes, Federico Frascoli, and James M McCaw. Periodic solutions in an sirws model with immune boosting and cross-immunity. *Journal of theoretical biology*, 410:55–64, 2016.

[14] Marc Lipsitch and Cécile Viboud. Influenza seasonality: lifting the fog. *Proceedings of the National Academy of Sciences*, 106(10):3645–3646, 2009.

[15] Pejman Rohani, Matthew J Keeling, and Bryan T Grenfell. The interplay between determinism and stochasticity in childhood diseases. *The American Naturalist*, 159(5):469–481, 2002.

[16] Asher Uziel and Lewi Stone. Determinants of periodicity in seasonally driven epidemics. *Journal of theoretical biology*, 305:88–95, 2012.

[17] World Health Organization. *The Global Burden of Disease: 2004 Update*. World Health Organization, Geneva, Switzerland, 2008. Data shows infectious and parasitic diseases caused approximately 9.5 million deaths (16.2% of global mortality) in 2004.